Long Quan Ton

HCMC, Vietnam | Email | LinkedIn | GitHub | Portfolio

Education

University of Technology Sydney & HCMUT — Ho Chi Minh City, Vietnam
*Bachelor of Artificial Intelligence; GPA: **6.75/7.0** (Dean's List 2026); IELTS 7.0* — *Expected 2028*

Experience

Ho Chi Minh City University of Technology (HCMUT) — Ho Chi Minh City
Undergraduate AI Research Assistant – AITechLab (ML4U) — *Aug 2026 – Present*

- Researching **Automatic Speech Recognition (ASR)** in the Speech Recognition Team under Dr. Nguyen Duc Dung, analyzing model internals and benchmarking STT/TTS/LALMs systems for low-latency, real-time conversational voice agents, contributing to upcoming scientific publications and technical reports.
- Benchmarking STT, TTS, and large audio-language models (LALMs) for low-latency conversational voice agents, and evaluating Small Language Models (SLMs) for on-device speech-to-speech pipelines.

FlyRank AI — Remote
AI Engineering Intern — *Jul 2026 – Oct 2026*

- Reduced new-workflow onboarding time by **60%** by designing a shared FastAPI and Redis layer that standardizes state, tool execution, and SQL access across LangGraph agents.
- Achieved **94%** valid text-to-SQL generation by building LangGraph multi-agent workflows on local LLMs (Ollama), with schema grounding and AST checks before execution.
- Cut mean time to resolve failed agent runs by **45%** by adding end-to-end request tracing and structured logging that ties every tool call to one trace ID.

Publications

AAC-BENCH: Benchmarking Audio as Context for Proactive Voice Agents — Under Review
*The Hieu Pham, Phuc Huynh, **Long Quan Ton**, Duc Dung Nguyen* — *AAMAS 2027 (CORE A*)*

Projects & Competitions

Weatherise – Vietnam AI Open Hackathon | Demo — NVIDIA/Viettel/Sovico
Project Lead & AI Developer — *Jun 2026*

- Led a 4-person team to a **Top 10 finish** with a multi-agent weather decision system for Tourism, Construction & Agriculture, matching domain experts on **91%** of go/no-go calls (**120 scenarios**) at **4.8s median latency** (**1,247 runs**) on **8**× NVIDIA H200, using LangGraph, NeMo Agent Toolkit, and NVIDIA NIM.
- Cut forecast error by **12% MAE** vs. the best single source (**22%** vs. raw GFS, **180 days**) by fusing **7 live weather APIs** through a **10-tool** MCP Server with bias correction, outlier rejection, and an LLM arbiter, using FastAPI and Redis 7.
- Achieved **0 unsafe go-calls** across **212 red-team prompts** (**94%** blocked at input, the rest vetoed by a deterministic rule engine) and **87.2% context recovery** via a **4-collection** RAG layer (∼**2,000 records**) behind NeMo Guardrails, using nv-embedqa-e5-v5, Qdrant, và PostgreSQL 16.

The Lantern – AssemblyAI Voice Agent Hackathon | Demo — AssemblyAI
Project Lead & AI Engineer — *Sep 2026*

- Led a 5-person team to build a real-time multilingual AI waiter (AssemblyAI, Qwen3 4B, Kokoro-82M, FastAPI) at under $1 per 1,000 orders; a "model interprets, code decides" pipeline raised six-turn order accuracy from 0/12 to **10/10**.
- Cut LLM p50 latency by **36.5%** (2.1s → 1.3s) and raised voice-session completion from 2/5 to 5/5 test sessions using ∼50ms fillers and barge-in over WebSocket streaming.

BKAi – Multi-Agent Admissions Counseling System — Personal Project
Full-Stack AI Developer — *Jan 2026 – Apr 2026*

- Built a LangGraph multi-agent RAG counselor with a deterministic verifier that answered **200/200** numeric admission questions exactly, by routing every score and quota to typed SQL tools over an 11-table fact DB and rejecting numbers absent from evidence.
- Lifted retrieval Hit@1 from 0.475 to **0.828** on 198 labeled queries by fusing Vietnamese dense and BM25 sparse vectors with RRF in Qdrant, plus cross-encoder reranking and parent–child chunking.
- Cut cold-answer p50 latency **15**× (27.3s → 1.79s) via a zero-LLM fast path, WebSocket streaming, and Gemini failover; shipped a Vietnamese voice agent at 1.26% CER with AssemblyAI STT and local Kokoro TTS.

Technical Skills

AI & Machine Learning: LangGraph, LangChain, MCP, RAG, AssemblyAI, PyTorch, Scikit-learn, XGBoost
Backend & Data: Python, FastAPI, PostgreSQL, SQLite, Redis, Qdrant
Frontend: TypeScript, React, Vite
Tools & Infrastructure: Docker, Git, WebSockets, vLLM, Ollama
Certifications: Generative AI with LLMs (AWS/DeepLearning.AI), Machine Learning Specialization (Stanford/DeepLearning.AI), AWS Cloud Practitioner Essentials (AWS)